

Food Waste Reduction Platform: Robust System Design and Project Management

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I. PROJECT MANAGEMENT DOCUMENTATION

A. Project Charter

Project Title:

Food Waste Reduction Platform: A Real-Time Geospatial Redistribution System.

Problem Statement and Justification:

Despite Colombia generating 9.7 million tons of food waste annually, local coordination gaps prevent the efficient redistribution of high-quality surplus within short freshness windows. This project bridges that gap by implementing a high-concurrency matching engine that prioritizes social equity and geographical proximity, directly supporting the circular economy objectives of the Bogotá university community.

1) SMART Objectives:

- **Specific:** To engineer and validate a mobile ecosystem connecting donor nodes (cafeterias/restaurants) with recipient nodes (students/charities) within a strictly enforced 3 km radius of the Carrera 8 #40-62 headquarters.
- **Measurable:** To achieve a Surplus Recovery Rate (SRR) of $\geq 80\%$, a backend algorithmic latency of < 2 seconds, and a Verified Pickup Rate (VPR) of $\geq 85\%$ under simulated load.
- **Achievable:** Leveraging the validated Workshop 2 stack: NestJS (logic), PostGIS (spatial), Redis (concurrency), and React Native (interface).
- **Relevant:** Directly addresses the 17 design requirements and 10 critical risks identified in the Workshop 1 systems diagnostic.
- **Time-bound:** Final system validation and performance benchmarking to be concluded by June 12, 2026.

2) *Comprehensive Team Structure and Accountability (RACI):* To ensure project success, the team operates under a RACI Matrix (Responsible, Accountable, Consulted, Informed), ensuring no ambiguity in deliverable ownership.

See Table I: Accountability Framework (RACI)

3) *Project Timeline and Critical Path:* The project follows a 16-week cycle divided into four major iterations. The Critical Path currently lies in the integration of the matching engine with real-time push notifications.

- **M1: System Diagnostic (Weeks 1-4):** Problem definition, IPO modeling, and boundary setting. [COMPLETED]

Table I: Accountability Framework (RACI)

Activity	Luna	Nico	Cris	Juan
Project Lead & Integration	A/R	I	I	I
Backend & Matching Engine	I	A/R	C	C
Frontend & UX Design	I	C	A/R	I
Quality Assurance (QA)	R	R	R	R

Legend: R: Responsible; A: Accountable; C: Consulted; I: Informed.

- **M2: Architectural Design (Weeks 5-8):** Component diagrams, technology stack selection, and risk mitigation planning. [COMPLETED]
- **M3: Development Optimization (Weeks 9-12):** Sprint-based coding, PostGIS implementation, and reliability scoring logic. [IN PROGRESS]
- **M4: Validation Final Report (Weeks 13-16):** Stress testing, experimental validation, and presentation of the "Best-in-Class" model.

4) *Resource Planning and Budgetary Considerations:* As an academic initiative, the budget is calculated in Human-Hours (HH) and Cloud Resource Credits:

- **Human Capital:** 4 Systems Engineers x 10 hours/week x 16 weeks = 640 total HH.
- **Technological Assets:** Expo/React Native (Open Source), PostgreSQL/PostGIS (Open Source), and Firebase (Free Tier).
- **Infrastructure Cost (Theoretical):** Estimated \$50 USD/month for production-grade hosting on AWS/Google Cloud for a pilot-scale deployment.

5) *Stakeholder Communication Strategy:* To maintain alignment with both internal and external stakeholders, the following communication matrix is enforced:

- **Technical Syncs (Internal):** Bi-weekly agile stand-ups to review GitHub issues and code reviews.
- **Academic Oversight:** Monthly progress reports submitted via the course platform to Eng. Carlos Andrés Sierra.
- **External Validation:** Semi-structured interviews with 3 local cafeteria managers to validate the "Same-day 2-4h window" feasibility.

6) *Scope and Constraints:* **Functional Boundary:** The system is restricted to digital coordination. It does not manage cold-chain

logistics, nor does it assume legal liability for food quality, which remains the donor's responsibility as per the system's "Terms of Use" architecture. **Assumptions:**

- 1) Continuous availability of GPS services for both donor and recipient.
- 2) Stable internet connectivity within the 3 km university radius.
- 3) Users (students/collectors) possess smartphones compatible with the Expo Go environment.

REFERENCES

- [1] Workshop 1 – Food Waste Reduction Mobile Application, A Rigorous System Analysis, Universidad Distrital Francisco José de Caldas, 2026.
- [2] Workshop 2 – System Design Document, Universidad Distrital Francisco José de Caldas, 2026.
- [3] IEEE Standards Association, "IEEE 730-2014 - IEEE Standard for Software Quality Assurance Processes," 2014.